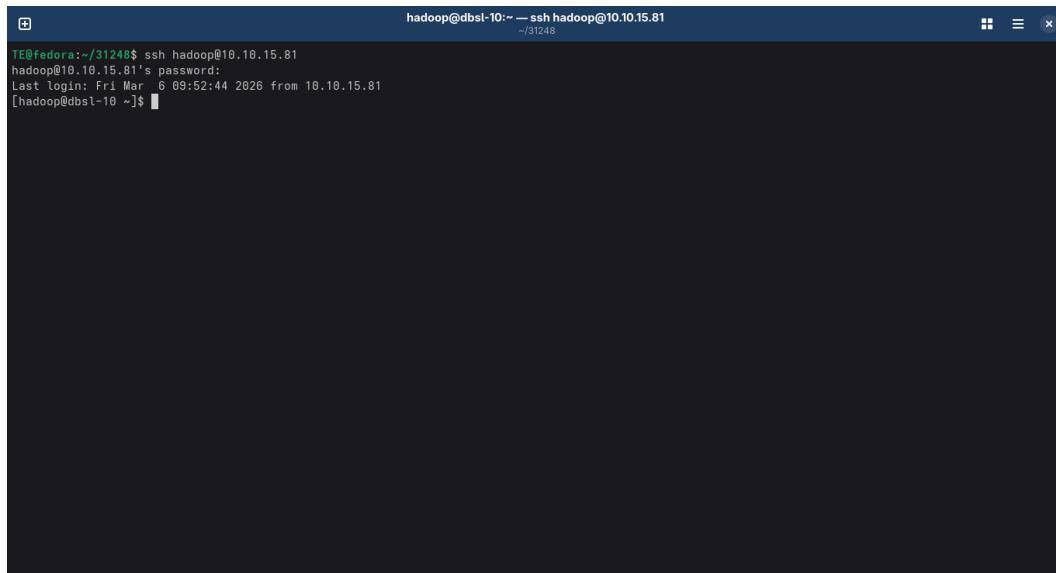


1. Connect to Server

ssh hadoop@10.10.15.81

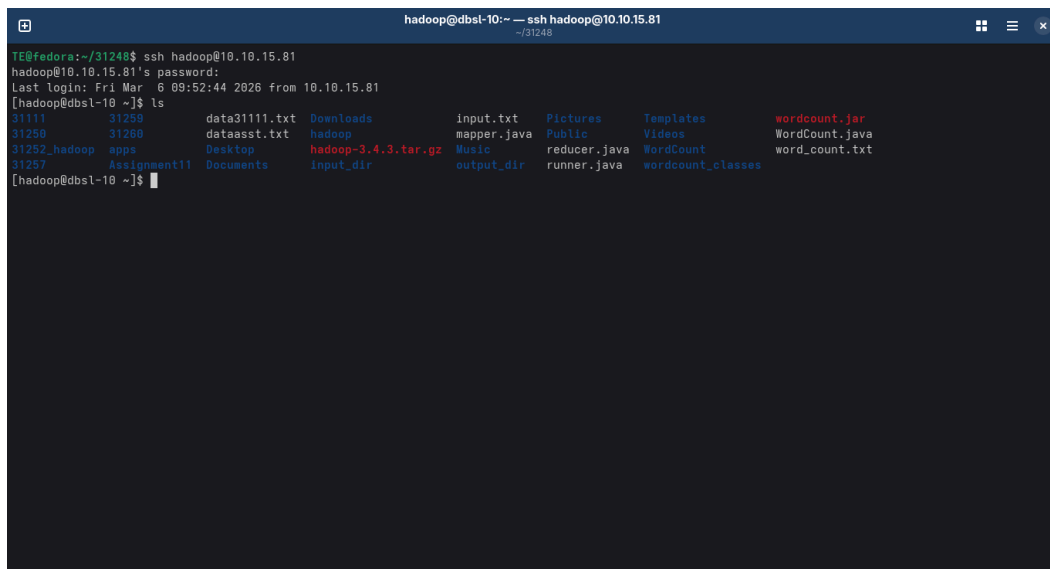
Enter password: hadoop



```
hadoop@dbstl-10:~ -- ssh hadoop@10.10.15.81
~/31248
TE@fedora:~/31248$ ssh hadoop@10.10.15.81
hadoop@10.10.15.81's password:
Last login: Fri Mar 6 09:52:44 2026 from 10.10.15.81
[hadoop@dbstl-10 ~]$
```

2. Check directories

ls



```
hadoop@dbstl-10:~ -- ssh hadoop@10.10.15.81
~/31248
TE@fedora:~/31248$ ssh hadoop@10.10.15.81
hadoop@10.10.15.81's password:
Last login: Fri Mar 6 09:52:44 2026 from 10.10.15.81
[hadoop@dbstl-10 ~]$ ls
31111      31258      data31111.txt  Downloads      input.txt      Pictures      Templates      wordcount.jar
31258      31260      dataasst.txt   hadoop          mapper.java    Public        Videos        WordCount.java
31252_hadoop  apps      Desktop        hadoop-3.4.3.tar.gz  Music          reducer.java  WordCount      word_count.txt
31257      Assignment11  Documents      input_dir       output_dir     runner.java   wordcount_classes

[hadoop@dbstl-10 ~]$
```

3. Change directory to: `cd $HADOOP_HOME/sbin`

```
31252_hadoop  apps      Desktop      hadoop-3.4.3.tar.gz  Music
31257        Assignment11 Documents    input_dir           output
[hadoop@dbsl-10 ~]$ cd $HADOOP_HOME/sbin
[hadoop@dbsl-10 sbin]$
```

4. Check directories in sbin folder : `ls`

```
[hadoop@dbsl-10 ~]$ cd $HADOOP_HOME/sbin
[hadoop@dbsl-10 sbin]$ ls
31248      hadoop-daemons.sh  start-all.sh      start-yarn.sh      stop-secure-dns.sh  WordCount.class
31255      https.sh           start-balancer.sh  stop-all.cmd      stop-yarn.cmd       wordcount_classes
data.txt   kms.sh             start-dfs.cmd      stop-all.sh       stop-yarn.sh        WordCount.jar
distribute-exclude.sh  mr-jobhistory-daemon.sh  start-dfs.sh      stop-balancer.sh   WordCount            workers.sh
FederationStateStore  refresh-namenodes.sh    start-secure-dns.sh  stop-dfs.cmd      'WordCount$IntSumReducer.class'  yarn-daemon.sh
hadoop-daemon.sh      start-all.cmd          start-yarn.cmd      stop-dfs.sh        'WordCount$TokenizerMapper.class'  yarn-daemons.sh
[hadoop@dbsl-10 sbin]$
```

5. Create a folder with your roll no:

```
[hadoop@dbsl-10 sbin]$ mkdir 31248
[hadoop@dbsl-10 sbin]$ ls
31248      hadoop-daemons.sh  start-all.sh      start-yarn.sh      stop-secure-dns.sh  WordCount.class
31255      https.sh           start-balancer.sh  stop-all.cmd      stop-yarn.cmd       wordcount_classes
data.txt   kms.sh             start-dfs.cmd      stop-all.sh       stop-yarn.sh        WordCount.jar
distribute-exclude.sh  mr-jobhistory-daemon.sh  start-dfs.sh      stop-balancer.sh   WordCount            workers.sh
FederationStateStore  refresh-namenodes.sh    start-secure-dns.sh  stop-dfs.cmd      'WordCount$IntSumReducer.class'  yarn-daemon.sh
hadoop-daemon.sh      start-all.cmd          start-yarn.cmd      stop-dfs.sh        'WordCount$TokenizerMapper.class'  yarn-daemons.sh
[hadoop@dbsl-10 sbin]$
```

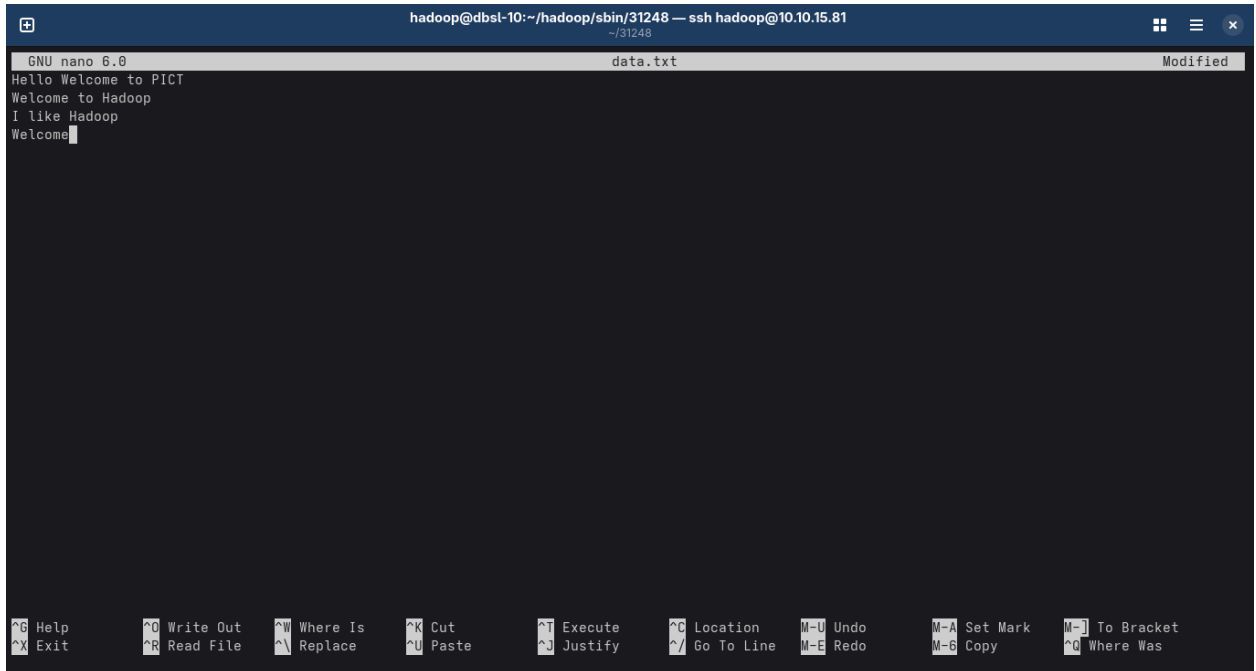
Then go to your working directory

```
data.txt      kms.sh           start-dfs.cmd      stop-all.sh      stop-yarn.sh      WordCount.jar
distribute-exclude.sh  mr-jobhistory-daemon.sh  start-dfs.sh      stop-balancer.sh  WordCount          workers.sh
FederationStateStore  refresh-namenodes.sh    start-secure-dns.sh  stop-dfs.cmd      'WordCount$IntSumReducer.class'  yarn-daemon.sh
hadoop-daemon.sh      start-all.cmd          start-yarn.cmd      stop-dfs.sh        'WordCount$TokenizerMapper.class'  yarn-daemons.sh
[hadoop@dbsl-10 sbin]$ cd 31248
[hadoop@dbsl-10 31248]$ ls
[hadoop@dbsl-10 31248]$
```

6. Create a text file

nano data.txt

Enter a text into it; This will be analysed by our MapReduce code (WordCount.java)



The screenshot shows a terminal window with a nano text editor. The title bar indicates the user is ssh'd into a hadoop machine at 10.10.15.81. The editor is editing a file named 'data.txt'. The content of the file is: 'Hello Welcome to PICT', 'Welcome to Hadoop', 'I like Hadoop', and 'Welcome'. The nano editor's status bar at the bottom shows various keyboard shortcuts like ^G Help, ^O Write Out, etc.

Then save it !

1. Ctrl + O
2. Enter
3. Ctrl + X



The screenshot shows a terminal window with the following commands and output: 'cd 31248', 'ls' (showing 'data.txt'), and 'nano data.txt'. The terminal also displays a table of Hadoop-related commands and their corresponding class names or scripts.

Command	Class/Script
FederationStateStore	refresh-namenodes.sh
hadoop-daemon.sh	start-all.cmd
start-secure-dns.sh	stop-dfs.cmd
start-yarn.cmd	stop-dfs.sh
'WordCount\$IntSumReducer.class'	yarn-daemon.sh
'WordCount\$TokenizerMapper.class'	yarn-daemons.sh

7. Create WordCount java program:

nano WordCount.java



The screenshot shows a terminal window with the following commands and output: 'cd 31248', 'ls' (showing 'data.txt'), and 'nano WordCount.java'. The terminal also displays a table of Hadoop-related commands and their corresponding class names or scripts.

Command	Class/Script
FederationStateStore	refresh-namenodes.sh
hadoop-daemon.sh	start-all.cmd
start-secure-dns.sh	stop-dfs.cmd
start-yarn.cmd	stop-dfs.sh
'WordCount\$IntSumReducer.class'	yarn-daemon.sh
'WordCount\$TokenizerMapper.class'	yarn-daemons.sh

Paste following code in the space:

```
import java.io.IOException;
import java.util.StringTokenizer;

import org.apache.hadoop.conf.Configuration;
import org.apache.hadoop.fs.Path;

import org.apache.hadoop.io.IntWritable;
import org.apache.hadoop.io.Text;

import org.apache.hadoop.mapreduce.Job;
import org.apache.hadoop.mapreduce.Mapper;
import org.apache.hadoop.mapreduce.Reducer;

import
org.apache.hadoop.mapreduce.lib.input.FileInputFormat;
import
org.apache.hadoop.mapreduce.lib.output.FileOutputFormat;

public class WordCount {

    public static class TokenizerMapper
        extends Mapper<Object, Text, Text, IntWritable>{

        private final static IntWritable one = new
IntWritable(1);
        private Text word = new Text();

        public void map(Object key, Text value, Context context)
            throws IOException, InterruptedException {

            StringTokenizer itr = new
StringTokenizer(value.toString());

            while (itr.hasMoreTokens()) {
                word.set(itr.nextToken());
                context.write(word, one);
            }
        }
    }
}
```

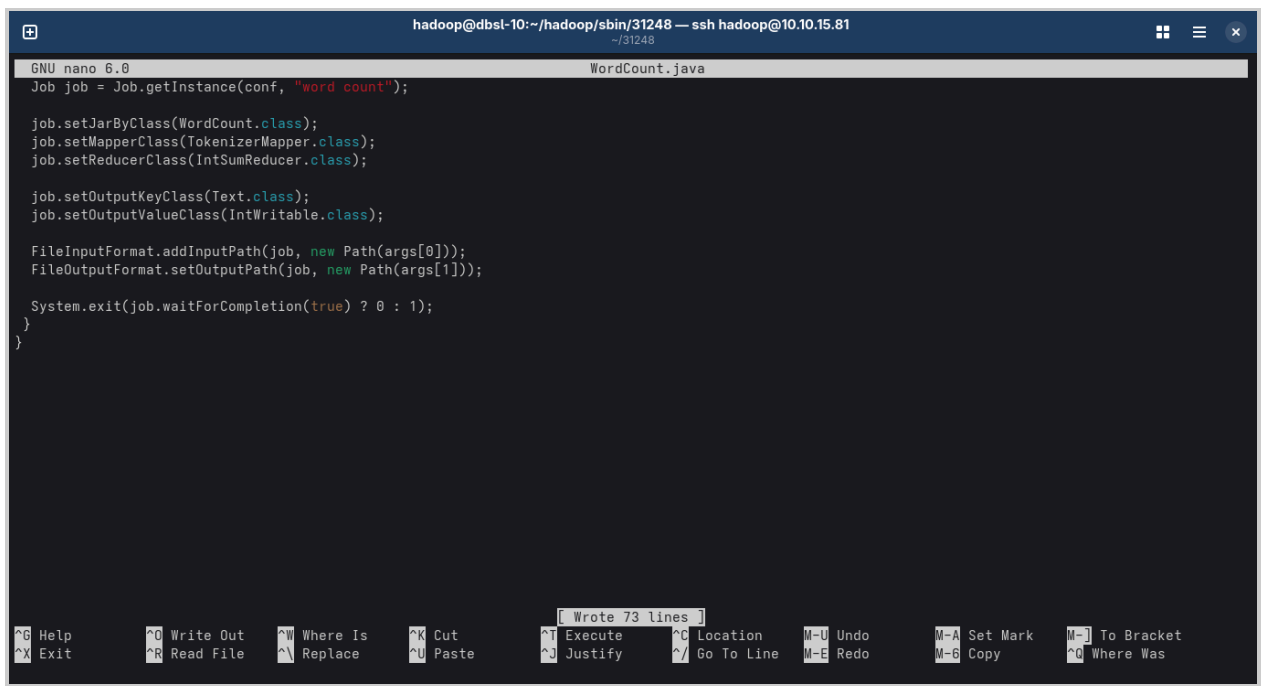
```
}  
}  
}  
  
public static class IntSumReducer  
    extends Reducer<Text,IntWritable,Text,IntWritable> {  
  
    private IntWritable result = new IntWritable();  
  
    public void reduce(Text key, Iterable<IntWritable>  
values,  
        Context context) throws IOException,  
InterruptedException {  
  
        int sum = 0;  
  
        for (IntWritable val : values) {  
            sum += val.get();  
        }  
  
        result.set(sum);  
        context.write(key, result);  
    }  
}  
  
public static void main(String[] args) throws Exception {  
  
    Configuration conf = new Configuration();  
    Job job = Job.getInstance(conf, "word count");  
  
    job.setJarByClass(WordCount.class);  
    job.setMapperClass(TokenMapper.class);  
    job.setReducerClass(IntSumReducer.class);  
  
    job.setOutputKeyClass(Text.class);  
    job.setOutputValueClass(IntWritable.class);  
}
```

```
FileInputFormat.addInputPath(job, new Path(args[0]));
FileOutputFormat.setOutputPath(job, new Path(args[1]));

System.exit(job.waitForCompletion(true) ? 0 : 1);
}
}
```

Then save it !

1. Ctrl + O
2. Enter
3. Ctrl + X

A screenshot of a terminal window with a dark blue title bar. The title bar text is 'hadoop@dbst-10:~/hadoop/sbin/31248 — ssh hadoop@10.10.15.81'. Below the title bar, the terminal shows the GNU nano 6.0 editor interface. The file being edited is 'WordCount.java'. The code in the editor matches the code block above. At the bottom of the terminal, there is a status bar with various keyboard shortcuts like ^G Help, ^O Write Out, ^W Where Is, ^X Cut, ^T Execute, ^C Location, M-U Undo, M-A Set Mark, M-J To Bracket, ^X Exit, ^R Read File, ^_ Replace, ^V Paste, ^J Justify, ^/ Go To Line, M-E Redo, M-G Copy, and ^Q Where Was. A small notification in the center of the status bar says '[Wrote 73 lines]'.

8. Now compile the java program:

```
javac -classpath `hadoop classpath` WordCount.java
```

Use this only if any error

```
javac -source 1.8 -target 1.8 -classpath `hadoop  
classpath` WordCount.java
```

```
[hadoop@dbsl-10 31248]$ nano data.txt
[hadoop@dbsl-10 31248]$ nano WordCount.java
[hadoop@dbsl-10 31248]$ ls
data.txt WordCount.java
[hadoop@dbsl-10 31248]$ javac -classpath `hadoop classpath` WordCount.java
[hadoop@dbsl-10 31248]$ ls
data.txt 'WordCount$IntSumReducer.class' 'WordCount$TokenizerMapper.class' WordCount.class WordCount.java
[hadoop@dbsl-10 31248]$
```

Now, create the .jar file:

```
jar cf WordCount.jar WordCount*.class
```

```
[hadoop@dbsl-10 31248]$ javac -classpath `hadoop classpath` WordCount.java
[hadoop@dbsl-10 31248]$ ls
data.txt 'WordCount$IntSumReducer.class' 'WordCount$TokenizerMapper.class' WordCount.class WordCount.java
[hadoop@dbsl-10 31248]$ jar cf WordCount.jar WordCount*.class
[hadoop@dbsl-10 31248]$ ls
data.txt 'WordCount$IntSumReducer.class' 'WordCount$TokenizerMapper.class' WordCount.class WordCount.jar WordCount.java
[hadoop@dbsl-10 31248]$
```

9. Now start the Hadoop services

A. Start HDFS: *start-dfs.sh*

```
hadoop@dbsl-10:~/hadoop/sbin/31248 — ssh hadoop@10.10.15.81
~/31248
[hadoop@dbsl-10 31248]$ start-dfs.sh
Starting namenodes on [localhost]
Starting datanodes
Starting secondary namenodes [dbsl-10]
2026-03-06 10:15:03,163 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
[hadoop@dbsl-10 31248]$
```

B. Start YARN: *start-yarn.sh*

```
ble
[hadoop@dbsl-10 31248]$ start-yarn.sh
Starting resourcemanager
resourcemanager is running as process 43335. Stop it first and ensure /tmp/hadoop-hadoop-resourcemanager.pid file is empty before retry.
Starting nodemanagers
localhost: nodemanager is running as process 43460. Stop it first and ensure /tmp/hadoop-hadoop-nodemanager.pid file is empty before retry.
[hadoop@dbsl-10 31248]$
```

C. Verify Hadoop Processes: *jps*

```
localhost: nodemanager is running as process 43460. Stop it first and ensure /tmp/hadoop-hadoop-nodemanager.pid file is empty before retry.
[hadoop@dbsl-10 31248]$ jps
43460 NodeManager
30628 NameNode
43335 ResourceManager
31609 SecondaryNameNode
31036 DataNode
51789 Jps
[hadoop@dbsl-10 31248]$
```

10. Now let's create HDFS directories:

A. Check the hdfs directory structure:

hdfs mkdir hdfs://localhost:9000/

```
[hadoop@dbsl-10 31248]$ hdfs ls hdfs://localhost:9000/
31248
31255_input
31255_output
input
input_31250
output
output_31259
tmp
user
[hadoop@dbsl-10 31248]$
```

To remove pre-existing dir:

hdfs rm -rf hdfs://localhost:9000/31248

B. Now create a folder with your roll no:

```
[hadoop@dbsl-10 31248]$ hdfs mkdir hdfs://localhost:9000/31248
[hadoop@dbsl-10 31248]$ hdfs ls hdfs://localhost:9000/
31248
31255_input
31255_output
input
input_31250
output
output_31250
output_31259
tmp
user
```

hdfs mkdir hdfs://localhost:9000/31248

C. Create the input file in your directory

```
user
[hadoop@dbsl-10 31248]$ hdfs mkdir hdfs://localhost:9000/31248/input
[hadoop@dbsl-10 31248]$ hdfs ls hdfs://localhost:9000/31248/
input
[hadoop@dbsl-10 31248]$
```

hdfs mkdir hdfs://localhost:9000/31248/input

11. Now upload input file (data.txt) to input dir

hdfs put data.txt hdfs://localhost:9000/31248/input


```
[hadoop@dbsl-10 31248]$ hdfs put data.txt hdfs://localhost:9000/31248/input
[hadoop@dbsl-10 31248]$ hdfs ls hdfs://localhost:9000/31248/input
data.txt
[hadoop@dbsl-10 31248]$
```

12. Run Wordcount MapReduce Job

*hadoop jar WordCount.jar WordCount /31248/input
/31248/output*

```
[hadoop@dbsl-10 31248]$ hadoop jar WordCount.jar WordCount /31248/input /31248/output
2026-03-06 10:32:48,266 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applica
ble
2026-03-06 10:32:48,866 INFO client.DefaultNoHARMAFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-03-06 10:32:49,459 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute
your application with ToolRunner to remedy this.
2026-03-06 10:32:49,502 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/hadoop/.staging/job_1772770996703_0011
2026-03-06 10:32:49,947 INFO input.FileInputFormat: Total input files to process : 1
2026-03-06 10:32:50,015 INFO mapreduce.JobSubmitter: number of splits:1
2026-03-06 10:32:50,325 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1772770996703_0011
2026-03-06 10:32:50,326 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-03-06 10:32:50,869 INFO conf.Configuration: resource-types.xml not found
2026-03-06 10:32:50,870 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-03-06 10:32:51,106 INFO impl.YarnClientImpl: Submitted application application_1772770996703_0011
2026-03-06 10:32:51,366 INFO mapreduce.Job: The url to track the job: http://dbsl-10:8088/proxy/application_1772770996703_0011/
2026-03-06 10:32:51,370 INFO mapreduce.Job: Running job: job_1772770996703_0011
```

```
2026-03-06 10:32:51,106 INFO impl.YarnClientImpl: Submitted application application_1772770996703_0011
2026-03-06 10:32:51,366 INFO mapreduce.Job: The url to track the job: http://dbsl-10:8088/proxy/application_1772770996703_0011/
2026-03-06 10:32:51,370 INFO mapreduce.Job: Running job: job_1772770996703_0011
2026-03-06 10:33:06,566 INFO mapreduce.Job: Job job_1772770996703_0011 running in uber mode : false
2026-03-06 10:33:06,567 INFO mapreduce.Job: map 0% reduce 0%
2026-03-06 10:33:11,638 INFO mapreduce.Job: map 100% reduce 0%
2026-03-06 10:33:16,675 INFO mapreduce.Job: map 100% reduce 100%
2026-03-06 10:33:16,695 INFO mapreduce.Job: Job job_1772770996703_0011 completed successfully
2026-03-06 10:33:16,832 INFO mapreduce.Job: Counters: 54
    File System Counters
        FILE: Number of bytes read=134
        FILE: Number of bytes written=619865
        FILE: Number of read operations=0
        FILE: Number of large read operations=0
        FILE: Number of write operations=0
        HDFS: Number of bytes read=169
        HDFS: Number of bytes written=50
        HDFS: Number of read operations=8
        HDFS: Number of large read operations=0
        HDFS: Number of write operations=2
        HDFS: Number of bytes read erasure-coded=0
    Job Counters
        Launched map tasks=1
        Launched reduce tasks=1
        Data-local map tasks=1
        Total time spent by all maps in occupied slots (ms)=2112
        Total time spent by all reduces in occupied slots (ms)=2283
        Total time spent by all map tasks (ms)=2112
        Total time spent by all reduce tasks (ms)=2283
        Total vcore-milliseconds taken by all map tasks=2112
        Total vcore-milliseconds taken by all reduce tasks=2283
        Total megabyte-milliseconds taken by all map tasks=2162688
        Total megabyte-milliseconds taken by all reduce tasks=2337792
```

```
Input split bytes=107
Combine input records=0
Combine output records=0
Reduce input groups=7
Reduce shuffle bytes=134
Reduce input records=11
Reduce output records=7
Spilled Records=22
Shuffled Maps =1
Failed Shuffles=0
Merged Map outputs=1
GC time elapsed (ms)=98
CPU time spent (ms)=980
Physical memory (bytes) snapshot=525197312
Virtual memory (bytes) snapshot=5093261312
Total committed heap usage (bytes)=434118464
Peak Map Physical memory (bytes)=310599680
Peak Map Virtual memory (bytes)=2542058944
Peak Reduce Physical memory (bytes)=214597632
Peak Reduce Virtual memory (bytes)=2550202368
Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
Bytes Read=62
File Output Format Counters
Bytes Written=50
[hadoop@dbsl-10 31248]$
```

13. Check Output directory

hdfs ls hdfs://localhost:9000/31248/output

```
Bytes Written=50
[hadoop@dbsl-10 31248]$ hdfs ls hdfs://localhost:9000/31248/output
_SUCCESS
part-r-000000
[hadoop@dbsl-10 31248]$
```

14. Display the Result

hdfs cat

hdfs://localhost:9000/31248/output/part-r-000000

```
part-r-000000
[hadoop@dbsl-10 31248]$ hdfs cat hdfs://localhost:9000/31248/output/part-r-000000
Hadoop 2
Hello 1
I 1
PICT 1
Welcome 3
like 1
to 2
[hadoop@dbsl-10 31248]$
```

15. Exit from HDFS Server

```
to 2  
[hadoop@dbsl-10 31248]$ exit  
logout  
Connection to 10.10.15.81 closed.  
TE@fedora:~/31248$
```